

L^AT_EX Author Guidelines for CVPR Proceedings

Anonymous CVPR submission

Paper ID *****

Abstract

001 The ABSTRACT is to be in fully justified italicized
002 text, at the top of the left-hand column, below the
003 author and affiliation information. Use the word “Ab-
004 stract” as the title, in 12-point Times, boldface type,
005 centered relative to the column, initially capitalized.
006 The abstract is to be in 10-point, single-spaced type.
007 Leave two blank lines after the Abstract, then begin
008 the main text. Look at previous CVPRabstracts to
009 get a feel for style and length.

010 1. Introduction

011 Please follow the steps outlined below when submitting
012 your manuscript to the IEEE Computer Society Press.
013 This style guide now has several important modifica-
014 tions (for example, you are no longer warned against
015 the use of sticky tape to attach your artwork to the
016 paper), so all authors should read this new version.

017 1.1. Language

018 All manuscripts must be in English.

019 1.2. Dual submission

020 Please refer to the author guidelines on the CVPR 2026
021 web page for a discussion of the policy on dual submis-
022 sions.

023 1.3. Paper length

024 Papers, excluding the references section, must be no
025 longer than eight pages in length. The references sec-
026 tion will not be included in the page count, and there
027 is no limit on the length of the references section. For
028 example, a paper of eight pages with two pages of ref-
029 erences would have a total length of 10 pages. There
030 will be no extra page charges for CVPR 2026.

031 Overlength papers will simply not be reviewed. This
032 includes papers where margins and formatting are
033 deemed to have been significantly altered from those
034 laid down by this style guide. Note that this L^AT_EX

guide already sets the figure captions and references in
a smaller font. The reason why such papers will not
be reviewed is that there is no provision for supervised
revisions of manuscripts. The review process cannot
determine the suitability of the paper for presentation
in eight pages if it is reviewed in 11 pages.

1.4. The ruler

The L^AT_EX style defines a printed ruler that should be
present in the version submitted for review. The ruler
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If you are preparing a document using a non-L^AT_EX
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Reviewers: note that the ruler measurements do not
align well with lines in the paper — this turns out to be
very difficult to do well when the paper contains many
figures and equations, and, when done, looks ugly. Use
fractional references (e.g., this line is 087.5), although
in most cases the approximate location would be ade-
quate.

1.5. Paper ID

Make sure that the Paper ID from the submission sys-
tem is visible in the version submitted for review (re-
placing the “*****” you see in this document). If you
are using the L^AT_EX template, make sure to update pa-
per ID in the appropriate place in the tex file.

1.6. Mathematics

Please, number all of your sections and displayed equa-
tions as in these examples:

$$E = m \cdot c^2 \quad (1)$$

and

$$v = a \cdot t. \quad (2)$$

073	It is important for the reader to be able to refer to any	explain the differences, just as you would if you had	121
074	particular equation. Just because you did not refer to	previously published related work. In such cases, in-	122
075	it in the text does not mean that some future reader	clude the anonymized parallel submission [12] as sup-	123
076	might not need to refer to it. It is cumbersome to have	plemental material and cite it as	124
077	to use circumlocutions like “the equation second from		
078	the top of page 3 column 1”. (Note that the ruler will	[1] Authors. “The frobnicatable foo filter”,	125
079	not be present in the final copy, so is not an alternative	F&G 2014 Submission ID 324, Supplied as	126
080	to equation numbers). All authors will benefit from	supplemental material fg324.pdf.	127
081	reading Mermin’s description of how to write mathe-		
082	matics: http://www.pamitc.org/documents/mermin.	Finally, you may feel you need to tell the reader that	128
083	pdf.	more details can be found elsewhere and refer them	129
084	1.7. Blind review	to a technical report. For conference submissions, the	130
085	Many authors misunderstand the concept of anonymiz-	paper must stand on its own, and not require the re-	131
086	ing for blind review. Blind review does not mean that	viewer to go to a tech report for further details. Thus,	132
087	one must remove citations to one’s own work—in fact	you may say in the body of the paper “further details	133
088	it is often impossible to review a paper unless the pre-	may be found in [13]”. Then submit the tech report as	134
089	vious citations are known and available.	supplemental material. Again, do not assume that the	135
090	Blind review means that you do not use the words	reviewers will read this material.	136
091	“my” or “our” when citing previous work. That is all.	Sometimes your paper is about a problem that you	137
092	(But see below for tech reports.)	tested using a tool that is widely known to be restricted	138
093	Saying “this builds on the work of Lucy Smith [1]”	to a single institution. For example, let’s say it’s 1969,	139
094	does not mean that you are Lucy Smith; it says that	you have solved a key problem on the Apollo lander,	140
095	you are building on her work. If you are Smith and	and you believe that the 1970 audience would like to	141
096	Jones, do not say “as we show in [7]”, say “as Smith	hear about your solution. The work is a development	142
097	and Jones show in [7]” and at the end of the paper,	of your celebrated 1968 paper entitled “Zero-g frobni-	143
098	include reference 7 as you would any other cited work.	cation: How being the only people in the world with	144
099	An example of a bad paper just asking to be re-	access to the Apollo lander source code makes us a wow	145
100	jected:	at parties”, by Zeus et al.	146
101	An analysis of the frobnicatable foo filter.	You can handle this paper like any other. Do not	147
102	In this paper, we present a performance anal-	write “We show how to improve our previous work	148
103	ysis of our previous paper [1], and show that	[Anonymous, 1968]. This time we tested the algorithm	149
104	it is inferior to all previously known methods.	on a lunar lander [name of lander removed for blind	150
105	Why the previous paper was accepted with-	review]”. That would be silly, and would immediately	151
106	out this analysis is beyond me.	identify the authors. Instead write the following:	152
107	[1] Removed for blind review	We describe a system for zero-g frobnication.	153
108	An example of an acceptable paper:	This system is new because it handles the fol-	154
109	An analysis of the frobnicatable foo filter.	lowing cases: A, B. Previous systems [Zeus	155
110	In this paper, we present a performance anal-	et al. 1968] did not handle case B properly.	156
111	ysis of the paper of Smith et al. [1], and show	Ours handles it by including a foo term in the	157
112	it to be inferior to all previously known meth-	bar integral.	158
113	ods. Why the previous paper was accepted	...	159
114	without this analysis is beyond me.	The proposed system was integrated with	160
115	[1] Smith, L and Jones, C. “The frobnicat-	the Apollo lunar lander, and went all the way	161
116	able foo filter, a fundamental contribution to	to the moon, don’t you know. It displayed the	162
117	human knowledge”. Nature 381(12), 1-213.	following behaviours, which show how well we	163
118	If you are making a submission to another confer-	solved cases A and B: ...	164
119	ence at the same time that covers similar or overlapping	As you can see, the above text follows standard sci-	165
120	material, you may need to refer to that submission to	entific convention, reads better than the first version,	166
		and does not explicitly name you as the authors. A re-	167
		viewer might think that it is likely that the new article	168
		was written by Zeus et al., but cannot make any deci-	169
		sion based on that guess. He or she would have to be	170

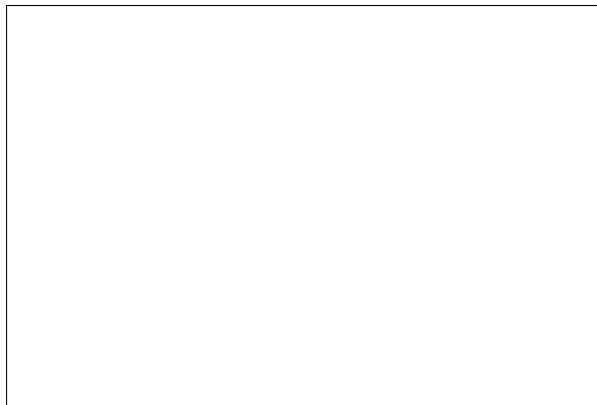


Figure 1. Example of caption. It is set in Roman so that mathematics (always set in Roman: $B \sin A = A \sin B$) may be included without an ugly clash.

sure that no other authors could have been contracted to solve problem B.

FAQ

Q: Are acknowledgements OK?

A: No. Leave them for the final copy.

Q: How do I cite my results reported in open challenges? A: To conform with the double-blind review policy, you can report results of other challenge participants together with your results in your paper. However, for your results, you should not identify yourself and should not mention your participation in the challenge. Instead, present your results referring to the method proposed in your paper and draw conclusions based on the experimental comparison with other results.

1.8. Miscellaneous

Compare the following:

`$conf_a$` *conf_a*

`$\mathit{conf}_a$` *conf_a*

See The TeXbook, p165.

The space after e.g., meaning “for example”, should not be a sentence-ending space. So e.g. is correct, e.g. is not. The provided `\eg` macro takes care of this.

When citing a multi-author paper, you may save space by using “et alia”, shortened to “et al.” (not “et. al.” as “et” is a complete word). If you use the `\etal` macro provided, then you need not worry about double periods when used at the end of a sentence as in Alpher et al. However, use it only when there are three or more authors. Thus, the following is correct: “Frobnication has been trendy lately. It was introduced by

Alpher [1], and subsequently developed by Alpher and Fotheringham-Smythe [2], and Alpher et al. [3].”

This is incorrect: “... subsequently developed by Alpher et al. [2] ...” because reference [2] has only two authors.

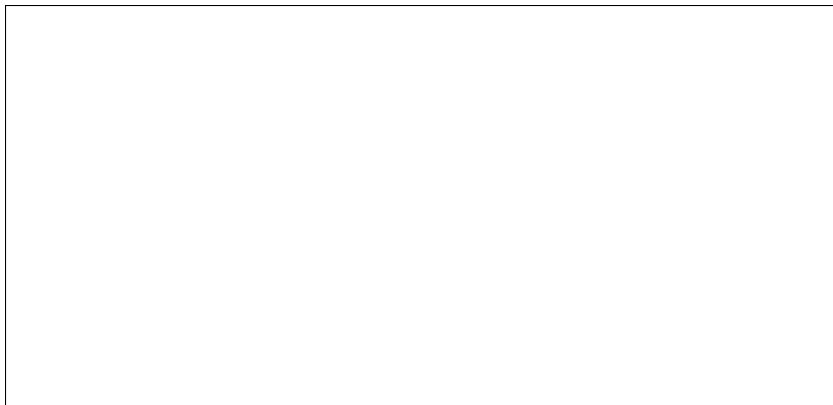
2. Related Work

2.1. Domain generalization for medical segmentation

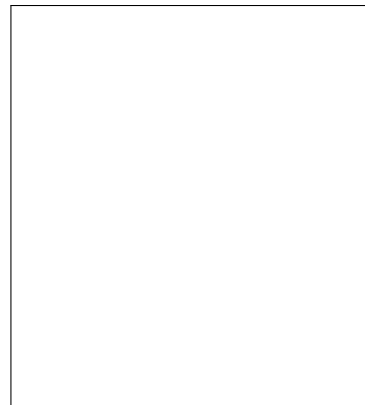
Domain generalization (DG) targets unseen distributions without access to target data, unlike unsupervised domain adaptation (UDA), which assumes unlabelled target images and solves an easier problem [23]. Most medical DG methods are multi-source, interpolating across several labelled domains by meta-learning [15], feature alignment [27], or frequency-space exchange [17]. Recent computer-vision entrants follow the same recipe: MADGNet couples a multi-frequency architecture with multi-domain training [19], and foundation models such as SynthFM pursue modality-agnostic segmentation at scale [22], while diffusion-based stylization normalizes appearance but relies on a generative prior and on source diversity [5]. We instead work in the single-source regime — one labelled dataset, no target data, no second source — which cannot exploit cross-domain interpolation and makes architecture-coupled methods orthogonal: our mechanism acts at the input level and leaves the segmentation network a fixed nnU-Net [10].

2.2. Synthesis-based contrast-agnostic training

The dominant single-source strategy is label-generative domain randomization, introduced by SynthSeg [6]: intensities are sampled from a per-label Gaussian mixture over an anatomical segmentation, yielding images of arbitrary contrast that force contrast-invariance. This family carries two structural costs. It is label-driven — a dense anatomical label map is required per subject, and SIAM shows the dependence is on label quality, needing a few high-grade manually curated templates [26]; producing such maps is expensive in expert hours. It is also texture-destroying: per-label Gaussian sampling replaces real gradients, partial-volume detail and unlabelled structure with piecewise noise, retaining layout but not texture. The lineage has been specialized to white-matter lesions [11] and MS lesions [7], and sits alongside training-free per-scan generative fitting such as SAMSEG [8]. We take SynthSeg’s label-free (EM) and original (noEM) variants as our primary baseline; as the EM variant has no public code, a faithful re-implementation is itself a contribution.



(a) An example of a subfigure.



(b) Another example of a subfigure.

Figure 2. Example of a short caption, which should be centered.

2.3. Single-source intensity and appearance augmentation

A second line manufactures appearance diversity directly from the real image, with no generative label model. General-purpose libraries stack intensity, spatial and artifact transforms on an nnU-Net trainer; we build on one such GPU framework [10, 18], which also serves as our augmentation-only ablation, and BigAug follows the same stacked-transform philosophy [29]. Closer to our operator are random-filter methods: GIN applies randomly re-initialized convolutions per forward pass to produce label-free appearance variants in true 3D with no trained parameters [20], RandConv establishes the principle for natural images [28], SLAug perturbs intensities per class using the task mask [24], and diffusion variants add learned generators [9]; a parallel frequency-domain line perturbs the Fourier amplitude spectrum rather than the image itself [16, 21, 30]. These share our label-free, single-source, architecture-agnostic stance — GIN especially is a direct peer — but vary appearance globally, spatially or spectrally, whereas our operator partitions the foreground into intensity classes and Voronoi sub-regions and applies a signed affine remap of the real intensities within each. The closest peer is semantic-aware random convolution [25], which also localizes the perturbation region-wise in true 3D, but differs on both axes that define our operator: it delineates regions from the ground-truth label map, reintroducing the task-label dependence we avoid, and applies a random convolution within each, which alters local gradients, whereas we partition label-free by k -means and Voronoi tessellation and apply an affine remap that leaves gradients intact. Because the remap is affine and region-local, edges and texture survive while contrast is randomized, preserving exactly the signal label-generative synthesis discards, without

any anatomical labels.

2.4. The evaluation gap

We are not aware of prior work reporting our protocol: training on a single real input modality and testing on held-out modalities of the same task and dataset, across a benchmark spanning brain, abdominal and tumour anatomy in both MRI and CT. Existing single-source evaluations are either cross-dataset but same-modality — e.g. FLAIR MS-lesion models across sites [4] on cohorts such as the public 3D-MR-MS data [14] — or contrast-agnostic-by-synthesis, where training data already span all contrasts and “single-modality” refers only to the test input. The one-real-modality-in, held-out-modality-out axis is the harder setting, and the one we target.

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